

Exercises Chapter 3

1. Consider the matrix

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

- (a) Determine $A^\top$. Is A symmetric?
- (b) Is A idempotent?
- (c) Compute the determinant and the rank. Is A nonsingular?
- (d) Compute the inverse.
- (e) Compute the trace.

2. (a) Let $AB = C$, where

$$A = \begin{pmatrix} 1 & 0 \\ 2 & 2 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix}.$$

Compute B .

- (b) Let δ and γ be $c \times 1$ vectors, X be a $d \times c$ matrix, and Y be a $c \times d$ matrix. Determine the orders of
- XY ,
 - YX ,
 - $\gamma^\top \gamma$,
 - $\gamma \gamma^\top$, and
 - $\delta^\top YX \gamma$.

Under which conditions do the expressions Y^{-1} and $\delta^\top YX + \gamma^\top \gamma$ exist?

- (c) Compute $\text{tr}(\lambda R^\top R)$ for $\lambda \in \mathbb{R}$ and

$$R = \begin{pmatrix} \frac{1}{4} & \frac{\sqrt{3}}{4} \\ \frac{\sqrt{3}}{4} & \frac{3}{4} \end{pmatrix}.$$

3. Let A be a nonsingular ($m \times m$) matrix. Simplify the expression as far as possible:

$$\left(\frac{1}{\sqrt{2}} A^{-1} \left(\frac{1}{\sqrt{2}} (A^\top)^\top + \frac{\sqrt{2}}{2} A \right) \right)^{-1}.$$

4. Consider the $n \times (p+1)$ matrix X with $\text{rank}(X) = p+1$. Moreover, let $P_X = X(X^\top X)^{-1}X^\top$, and let $M_X = I_n - P_X$

- (a) Determine the order of the following matrices:

- I_n
- $X^\top X$
- P_X
- M_X

- (b) Which matrices from (a) are symmetric?
- (c) Which matrices from (a) are idempotent?
- (d) Compute the trace of I_n and P_X .

5. Let X be a $n \times (p+1)$ matrix. Show that $X^\top X$ is positive semi-definite. Under which condition is $X^\top X$ positive definite?

6. Let $y \in \mathbb{R}^n$, X be a $n \times (p+1)$ matrix, $b \in \mathbb{R}^{(p+1)}$ and let

$$f(b) = (y - Xb)^\top (y - Xb).$$

Compute the following derivatives of $f(b)$ with respect to b :

(a)

$$\frac{\partial f(b)}{\partial b},$$

(b)

$$\frac{\partial^2 f(b)}{\partial b \partial b^\top},$$